

Load Interconnection System Impact Study Report

For Queue Project Q11-0039

Vonore 345 kV

May 03, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0039. The Interconnection Customer proposes a 357.2 MW large load (data center) facility interconnecting at the Vonore 345 kV substation in the West Virginia area. The request was received on May 24, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 32 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$38,358,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: Excell to Vasona 138kV
Contingency: Vonore to Morgantown Energy Facility 345kV
Loading: 205.0% (Rating: 198 MVA)
DFAX: 100.0%
Pre-project loading: 24.5%
2. Facility: Oakely to Vasona 138kV
Contingency: Vonore to Morgantown Energy Facility 345kV
Loading: 199.5% (Rating: 202 MVA)
DFAX: 100.0%
Pre-project loading: 22.8%
3. Facility: W Murray to Okely 138kV
Contingency: Vonore to Morgantown Energy Facility 345kV line
Loading: 194.9% (Rating: 206 MVA)
DFAX: 100.0%
Pre-project loading: 21.3%
4. Facility: W Murray to Titans Substation 138kV
Contingency: Vonore to Morgantown Energy Facility 345kV
Loading: 191.6% (Rating: 208 MVA)
DFAX: 100.0%
Pre-project loading: 19.9%
5. Facility: Titans to Vonore 138kV
Contingency: Morgantown Energy Facility to Vonore 345kV L
Loading: 165.0% (Rating: 229 MVA)
DFAX: 100.0%
Pre-project loading: 9.2%
6. Facility: Excell to Coalinga 1 138kV
Contingency: Vonore to Morgantown Energy Facility 345kV
Loading: 139.3% (Rating: 181 MVA)

DFAX: 62.9%
Pre-project loading: 15.4%

7. Facility: Vonore 138/1kV Autotransformer
Contingency: Vonore Substation to Morgantown Energy Facility 345kV
Loading: 116.9% (Rating: 323 MVA)
DFAX: 100.0%
Pre-project loading: 6.5%
8. Facility: Vonore 345 kV/1 kV Transformer Bank
Contingency: Vonore Substation to Morgantown Energy Facility 345kV
Loading: 108.0% (Rating: 349 MVA)
DFAX: 100.0%
Pre-project loading: 5.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Excell to Vasona 138kV	Rebuild line with higher-capacity conductor	\$2,361,000	100% (\$2,361,000)
Oakely to Vasona 138kV line	Rebuild line with higher-capacity conductor	\$2,310,000	100% (\$2,310,000)
West Murray to Oakely 138kV line	Rebuild line with higher-capacity conductor	\$4,290,000	100% (\$4,290,000)
W Murray to Titans 138kV line	Rebuild line with higher-capacity conductor	\$2,193,000	100% (\$2,193,000)
Titans to Vonore 138kV	Rebuild line with higher-capacity conductor	\$3,955,000	100% (\$3,955,000)
Excell to Coalinga 1 138kV line	Rebuild line with higher-capacity conductor	\$6,208,000	100% (\$6,208,000)
Vonore 138/1kV XFMR	Replace transformer with higher-rated unit	\$3,618,000	100% (\$3,618,000)
Vonore 345/1kV TX	Replace transformer with higher-rated unit	\$13,423,000	100% (\$13,423,000)
	Total allocated to project		\$38,358,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.